



# Compact polarization-independent quasi-adiabatic $2 \times 2$ 3 dB coupler on silicon

HUNG-CHING CHUNG,<sup>1</sup> CHIH-HSIEN CHEN,<sup>2</sup> YUNG-JR HUNG,<sup>2,3</sup>  
AND SHUO-YEN TSENG<sup>1,4</sup> 

<sup>1</sup>Department of Photonics, National Cheng Kung University, 1 University Rd., Tainan 701, Taiwan

<sup>2</sup>Department of Photonics, National Sun Yat-sen University, 70 Lien hai Rd., Kaohsiung 804, Taiwan

<sup>3</sup>yungrjr@mail.nsysu.edu.tw

<sup>4</sup>tsengsy@mail.ncku.edu.tw

**Abstract:** We demonstrate a quasi-adiabatic polarization-independent  $2 \times 2$  3 dB coupler based on the silicon-on-insulator platform. Using a quasi-adiabatic taper design for the mode evolution/coupling region, the TE mode evolution is accelerated, and the TM mode coupling is achieved at a short coupling length. The measured working bandwidth is 75 nm with a compact mode evolution/coupling region of 11.7  $\mu\text{m}$ .

© 2022 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

## 1. Introduction

Silicon-on-insulator (SOI) is a promising platform for integrated photonics due to its compatibility with the existing complementary metal-oxide-semiconductor (CMOS) fabrication technology and the high density of integration enabled by large refractive index contrast. On the other hand, the large refractive index contrast and non-square geometry of typical SOI waveguides also give rise to strong birefringence of the waveguides, resulting in strong polarization dependence of the devices on SOI. Two-times-two ( $2 \times 2$ ) 3 dB couplers are one of the crucial building blocks in SOI-based photonic integrated circuits and are often used in switches [1] and multiplexers [2]. However, the performances of couplers based on conventional directional couplers (DCs) are highly sensitive to the polarization and wavelength [3]. Several approaches have been proposed to realize polarization-independent and broadband 3 dB couplers, including bent DCs [4], multimode interference couplers (MMIs) [5–7], Y-junctions [8], and adiabatic couplers (ADCs) [9]. Bent DCs achieve polarization independence by cascading two DCs with different bending radii designed for different polarizations, but they have a large footprint and high sensitivity to fabrication variations. MMIs and Y-junctions can achieve small footprint and low loss, but they typically find applications as  $1 \times 2$  power splitters. ADCs have large bandwidth and good fabrication tolerance, but they are hampered by large device footprint. Subwavelength-grating (SWG) assisted devices [10,11] have been recently demonstrated to exhibit polarization-independent and broadband  $2 \times 2$  3 dB couplers; but SWG-based devices are sensitive to grating pitch and fill factor variations, and it may be challenging to meet the required fabrication precision using standard CMOS-compatible optical lithography.

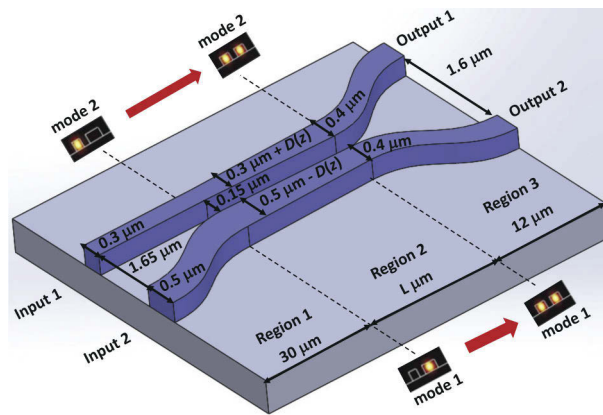
The operating principle of  $2 \times 2$  ADCs involves the evolution of individual modes. By optimizing the mode evolution region for both the transverse electric (TE) modes and the transverse magnetic (TM) modes, ADCs can be made polarization-independent while maintaining large operating bandwidth and fabrication tolerance [9,12]. But long mode evolution regions are needed to satisfy the adiabatic criteria for both polarizations. In a typical silicon  $2 \times 2$  3 dB ADC, the TE mode evolution occurs over a much longer length than the TM mode due to stronger confinement of the TE mode [9]. As a result, evolution of the TE mode presents the real bottleneck in reducing the device length of polarization-independent ADCs.

There have been numerous approaches to optimize the design of adiabatic devices in integrated optics in order to reduce the device length [13–17]. It has been shown that the fast quasi-adiabatic

(FAQUAD) protocol in quantum control [18,19] can effectively accelerate the slow mode evolution in adiabatic waveguides [15,20]. In this paper, we speed up the TE mode evolution by designing the 3 dB coupler in the quasi-adiabatic regime, resulting in a compact and broadband design. At the same time, when the device is short and the TM mode no longer satisfies the adiabatic criterion, the weaker confinement of the TM mode means that the TM mode can have a very short coupling length under mode coupling operation (rather than mode evolution). It is well known in mode coupling devices that the coupler bandwidth is inversely proportional to the coupler length [21]. Moreover, mode coupling in mismatched waveguides can also desensitize the coupler to variations [22]. In this quasi-adiabatic regime, the TM mode works under mode coupling, with a large bandwidth resulting from the short coupling region and mismatched waveguides. The result is a compact polarization-independent 2×2 3 dB quasi-adiabatic coupler (QADC) on the SOI platform. The coupler has a short mode evolution/coupling region of 11.7  $\mu\text{m}$ . Devices fabricated using multiple project wafer (MPW) foundry service show polarization-independent transmission for a bandwidth of 75nm (1490-1565 nm).

## 2. Design and simulation

A schematic for our coupler is shown in Fig. 1. The coupler is based on SOI strip waveguides consisting of 2  $\mu\text{m}$  thick buried oxide layer, 220 nm silicon strip, and covered with silica cladding. The device is divided into three regions as shown in the figure. In the 30  $\mu\text{m}$  long Region 1, two waveguides with initial widths  $W_1=300$  nm and  $W_2=500$  nm are brought together using an S-bend to reduce the gap from 1.65  $\mu\text{m}$  to 150 nm. The key element of this design is the mode evolution/coupling region (Region 2) of length  $L$ . A smaller gap between the waveguides can reduce the mode evolution region, and we choose a constant gap of 150 nm as limited by the minimum feature size of the foundry service. A taper function  $D(z)$  is applied to both waveguides so their widths becomes  $W_1 + D(z)$  and  $W_2 - D(z)$ . With the boundary conditions,  $D(0) = 0$ ,  $D(L) = 100$  nm, the dissimilar waveguides are converted to two identical waveguides of the same width,  $W_1 = W_2 = 400$  nm, at the end of Region 2. In the 12  $\mu\text{m}$  long Region 3, two s-bends separate the waveguide gap from 150 nm to 1.6  $\mu\text{m}$ . In this work, we design the taper function  $D(z)$  so that the TE mode operates by mode evolution and the TM mode operates by mode coupling to achieve polarization-independent 3 dB coupling. Next, we discuss the different operating regimes of the coupler by analyzing device adiabaticity.



**Fig. 1.** Schematic of the quasi-adiabatic coupler (QADC). Insets show the supermodes of the coupler at the beginning and the end of Region 2.

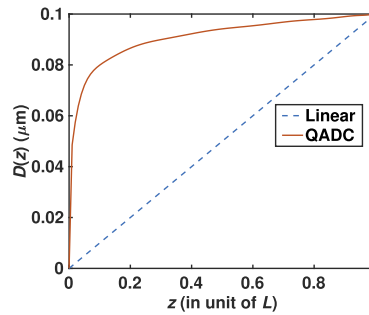
### 2.1. Adiabaticity analysis of the conventional ADC

In conventional ADCs, when TE or TM light is launched into input 1 or input 2, only one of the supermodes (local eigenmodes) (shown by the insets in Fig. 1) is excited initially in Region 2. Due to waveguide asymmetry, the supermodes match the guided modes of the individual input waveguides. As the excited mode adiabatically propagates along the coupler, very little or no power is coupled into the other supermode of the same polarization. At the output, the initially excited supermode evolves into the even/odd supermode of the two identical waveguides, and the power is split evenly. It is key to ensure that mode evolution in Region 2 is done slowly (adiabatically) enough to minimize unwanted coupling into the other supermode. If the adiabaticity criterion is satisfied for both TE and TM polarization modes, the device functions as a polarization-independent 3 dB coupler.

To analyze device adiabaticity during evolution, we can calculate the adiabaticity parameters along the propagation direction. The adiabaticity parameter for vectorial field propagating in the  $z$  direction in an optical waveguide can be defined as [23]

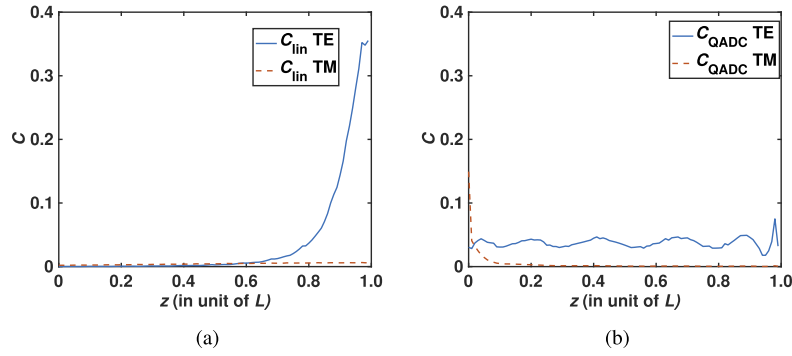
$$c(z) = \left| \frac{\int [\mathbf{E}_{mt} \times \frac{\partial}{\partial z} \mathbf{H}_{nt}^*] \cdot \hat{\mathbf{z}} dS}{\beta_m - \beta_n} \right| = \left| \frac{\langle m | \dot{n} \rangle}{\beta_m - \beta_n} \right|, \quad (1)$$

where  $\mathbf{E}_{mt}$  and  $\mathbf{H}_{nt}$  are the transverse components of the electric field and magnetic field associated with the  $m$ th and  $n$ th mode of the optical waveguide,  $S$  is the entire waveguide cross section, and the dot denotes derivative with respect to  $z$ . This definition is indicative of the coupling strength between modes  $|m\rangle$  and  $|n\rangle$ . For light evolution to follow a single supermode  $|m\rangle$  in an adiabatic device, it is important to make  $c \ll 1$  anywhere during the evolution. The adiabaticity parameter along any point of the mode evolution region can be obtained using Eq. (1) with mode data calculated using a mode solver. In conventional designs [9], the waveguides in region 2 are tapered according to a linear function as shown by the dashed line in Fig. 2. Using (1), we obtain the adiabaticity parameters of the conventional linear taper structure for both TE and TM polarizations as shown in Fig. 3(a), where  $c$  scales inversely with  $L$ . In order to satisfy the adiabatic criterion ( $c \ll 1$ ) anywhere during evolution, the length  $L$  would need to be long enough to keep the overall adiabaticity parameter below an acceptable level. We also note that the total adiabaticity parameter of the TE mode is larger than that of the TM mode, as a result, we expect the TE mode evolution to occur over a longer length.



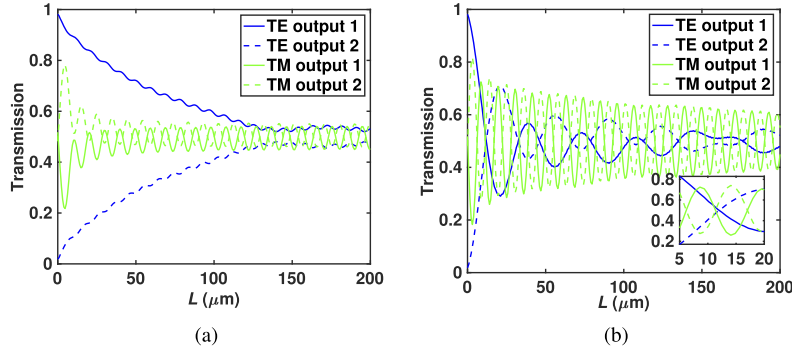
**Fig. 2.** Plot of the taper functions  $D(z)$ . Dashed line: conventional linear taper. Solid line: quasi-adiabatic taper.

Using a commercial software (FIMMPROP, Photon Design) employing a full vectorial eigenmode expansion method (EME), we can simulate light propagation in the couplers. In Fig. 4(a) we plot the coupler outputs as a function of  $L$  by exciting input 1 at  $\lambda = 1.55 \mu\text{m}$  for both polarizations. The coupler functions as a 3 dB coupler when the output curves are close. It is



**Fig. 3.** Adiabaticity parameters along the propagation direction in region 2 for the TE and the TM polarizations. (a) Conventional linear taper function. (b) Quasi-adiabatic taper function.

clear that the TE mode need much longer evolution length to evenly distribute its power between the outputs ( $L > 130 \mu\text{m}$ ), while the TM mode achieves its first crossing at  $L = 20 \mu\text{m}$ . This result agrees with the previous adiabaticity analysis showing larger adiabaticity parameter for the TE mode. The TE mode evolution thus limits the length of ADCs.



**Fig. 4.** Outputs of the couplers as a function of  $L$  for the TE and the TM mode inputs. (a) Conventional linear ADC. (b) Quasi-adiabatic coupler. Inset: Magnified view of the region around  $L = 11.7 \mu\text{m}$ .

## 2.2. Adiabaticity analysis of the quasi-adiabatic coupler (QADC)

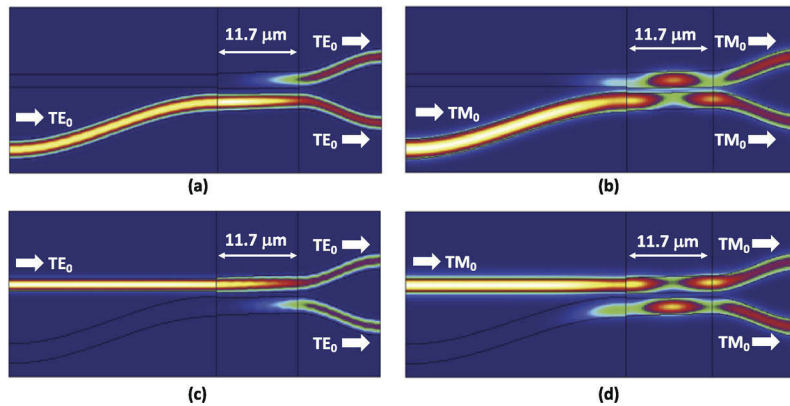
Due to the overall larger adiabaticity parameter of the TE mode, we focus first on reducing its mode evolution length. The FAQUAD protocol achieves shortcuts to adiabaticity by redistributing adiabaticity parameter homogeneously along the propagation length. Following the procedures listed in [20], we obtain the quasi-adiabatic taper function shown by the solid line in Fig. 2. Using (1), we obtain the adiabaticity parameters of the quasi-adiabatic taper structure for both TE and TM polarizations as shown in Fig. 3(b). We can see that the adiabaticity parameter of the TE mode is now homogenized, however, the fast-changing  $D(z)$  at the beginning leads to large adiabaticity parameter of the TM mode at the beginning of Region 2. The large adiabaticity parameter of the TM mode indicates that it no longer operates under mode evolution, but coupling between the TM eigenmodes is expected.

The simulated QADC outputs as a function of  $L$  at  $\lambda = 1.55 \mu\text{m}$  when input 1 is excited are shown in Fig. 4(b) for both polarizations. Compared with Fig. 4(a), we can see that the TE

mode now reaches its first crossing at a short length of  $L = 11.7 \mu\text{m}$ . So, the mode evolution length of the TE mode has been successfully reduced in the QADC. At the same time, the TM mode exhibits larger oscillations with  $L$  that are indicative of mode coupling operation in the QADC, while the smaller oscillations in Fig. 4(a) indicate mode evolution operation of the TM mode in conventional linear ADC. Compared with conventional DCs consisting of identical waveguides that has 100% power oscillations between waveguides, the TM mode shows a swing of around 70% in the QADC consisting of dissimilar waveguides. Similar to bent DCs [22] where a mismatch between coupled waveguides can desensitize the coupler to variations, the TM mode in the QADC should have less sensitivity to parameter variations than in conventional DCs. Close to the TE mode crossing at  $L=11.7 \mu\text{m}$ , the TM mode curves also have a crossing, indicating polarization-independent 3 dB coupling can be achieved for both polarizations at a short length under this quasi-adiabatic regime.

### 2.3. Simulation of the quasi-adiabatic coupler (QADC)

For the  $L=11.7 \mu\text{m}$  3 dB QADC, we show the simulated light propagation in the device by exciting both input ports for both polarizations at  $1.55 \mu\text{m}$  input wavelength in Fig. 5. Clearly, the input light is evenly split into the two output ports for both the TE and the TM modes using both input ports. From the light distributions, we can see that the TE mode operates under mode evolution, the input supermodes evolve to the even and odd supermodes at the end of Region 2. For the TM mode, light travels back and forth between the waveguides indicating the expected mode coupling operation. At this point, we can see that polarization-independent 3 dB coupling is achieved in the QADC with a short  $11.7 \mu\text{m}$  mode evolution/coupling region.

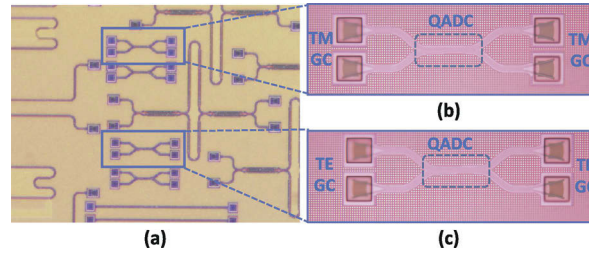


**Fig. 5.** EME simulated light distributions in the polarization-independent quasi-adiabatic 3-dB coupler with an  $L=11.7 \mu\text{m}$ . Input 2: (a) TE mode (b) TM mode. Input 1: (c) TE mode (d) TM mode.

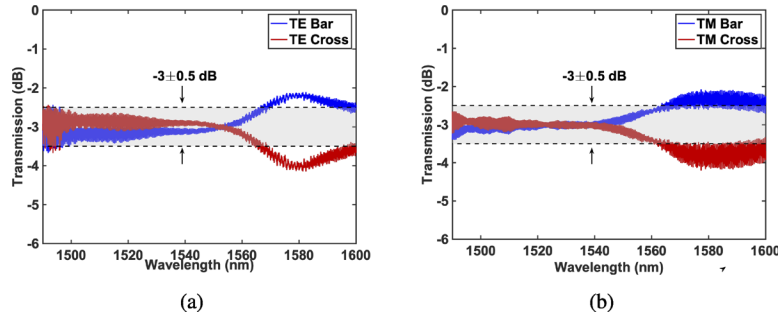
## 3. Fabrication and experimental results

Through a MPW foundry service, the QADC 3 dB couplers were fabricated on an 8-inch silicon-on-insulator wafer with 220 nm crystalline silicon and  $3 \mu\text{m}$  oxide top and bottom cladding using a CMOS-compatible process with ArF 193-nm deep ultraviolet lithography. Optical microscope images of the fabricated devices are shown in Fig. 6. The spectral responses of all fabricated devices were probed using a custom-built interrogation system and characterized using a tunable laser with a wavelength tuning range of 1480 to 1620 nm and a step resolution of 1 pm. A pair of grating couplers with a coupling efficiency of -4.9/-5.2 dB for the TE [Fig. 6(c)] and the TM [Fig. 6(b)] polarizations were utilized for optical input/output between single mode





**Fig. 6.** (a) Optical microscope image of the fabricated QADC 3-dB couplers on SOI. Magnified view of the couplers with (b) TM and (c) TE grating couplers (GCs) attached to the input and output ports.



**Fig. 7.** Measured transmission spectra of the QADC 3 dB coupler. (a) TE mode. (b) TM mode.

fiber and silicon chip. The proximity of these two QADCs ( $400\ \mu\text{m}$ ) on the wafer should ensure that the devices are almost identical and thus have identical performance for each polarization [24]. Figure 7 shows the measured transmission spectra at bar and cross ports of the QADC 3 dB couplers for the TE and the TM polarizations. The spectra are normalized using the procedures outlined in [25] to eliminate the impact of fiber coupling variation to the extracted power splitting ratio. For the TE polarization, the bar and cross transmission curves cross at close to the designed  $1550\ \text{nm}$ , and the larger imbalance towards the longer wavelength is consistent with the previously demonstrated FAQUAD splitter [20]. The large operating bandwidth of the TE mode is a result of the accelerated mode evolution in the QADC. For the TM polarization, the curves cross at a shorter wavelength close to  $1540\ \text{nm}$  because the perfect 3 dB coupling length for the TM mode happens at a slightly shorter length than the designed  $L=11.7\ \mu\text{m}$  as shown in Fig. 4(b). TM mode coupling between dissimilar waveguides result in the observed large operating bandwidth. From the measurement results, we can infer that the QADC profile in Fig. 2 was closely realized in fabrication. Otherwise, deviation from the QADC profile should result in variations of the device adiabaticity parameter [Fig. 3(b)], which in turn leads to variations of the output characteristics shown in Fig. 4(b), resulting in deviation of the curve crossings from the design in Fig. 7. From the figures, the polarization independent  $3\pm 0.5\ \text{dB}$  bandwidth is about  $75\ \text{nm}$  ( $1490\text{--}1565\ \text{nm}$ ), limited by the bandwidth of the tunable laser and grating couplers. The measurement results show that the QADC indeed achieves polarization-independent 3 dB coupling over a broad bandwidth with a short evolution/coupling length.

#### 4. Conclusion

In conclusion, we have proposed and demonstrated a broadband and polarization-independent quasi-adiabatic  $2\times 2$  3 dB coupler on SOI. Short mode evolution length for the TE mode is

achieved using the FAQUAD protocol, while the TM mode has a short mode coupling length. The measurement results show that the QADC have a splitting ratio varying between 45%: 55% to 55%: 45% in the wavelength range from 1490 nm to 1565 nm for both TE and TM polarizations. The device is compatible with common silicon photonics foundry processes.

**Funding.** Ministry of Science and Technology, Taiwan (108-2221-E-006-204-MY3, 109-2221-E-110-068-MY3, 110-2224-E-992-001).

**Acknowledgments.** The authors would like to acknowledge chip fabrication support provided by Taiwan Semiconductor Research Institute (TSRI).

**Disclosures.** The authors declare no conflicts of interest.

**Data availability.** Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

## References

1. J. V. Campenhout, W. M. J. Green, S. Assefa, and Y. A. Vlasov, "Low-power, 2×2 silicon electro-optic switch with 110-nm bandwidth for broadband reconfigurable optical networks," *Opt. Express* **17**(26), 24020–24029 (2009).
2. F. Horst, W. M. J. Green, S. Assefa, S. M. Shank, Y. A. Vlasov, and B. J. Offrein, "Cascaded Mach-Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-)multiplexing," *Opt. Express* **21**(10), 11652–11658 (2013).
3. H. Yamada, T. Chu, S. Ishida, and Y. Arakawa, "Optical directional coupler based on Si-wire waveguides," *IEEE Photonics Technol. Lett.* **17**(3), 585–587 (2005).
4. X. Chen, W. Liu, Y. Zhang, and Y. Shi, "Polarization-insensitive broadband 2×2 3 dB power splitter based on silicon-bent directional couplers," *Opt. Lett.* **42**(19), 3738–3740 (2017).
5. D. Dai and S. He, "Proposal for diminishment of the polarization-dependency in a Si-nanowire multimode interference (MMI) coupler by tapering the MMI section," *IEEE Photonics Technol. Lett.* **20**(8), 599–601 (2008).
6. Z. Xiao, X. Luo, P. H. Lim, P. Prabhathan, S. T. H. Silalahi, T.-Y. Liow, J. Zhang, and F. Luan, "Ultra-compact low loss polarization insensitive silicon waveguide splitter," *Opt. Express* **21**(14), 16331–16336 (2013).
7. D. P. Nair and M. Ménard, "A compact low-loss broadband polarization independent silicon 50/50 splitter," *IEEE Photonics J.* **13**(4), 1–7 (2021).
8. Y. Wang, S. Gao, K. Wang, and E. Skafidas, "Ultra-broadband and low-loss 3 dB optical power splitter based on adiabatic tapered silicon waveguides," *Opt. Lett.* **41**(9), 2053–2056 (2016).
9. Y. Wang, L. Xu, H. Yun, M. Ma, A. Kumar, E. El-Fiky, R. Li, N. Abadifacalvo, L. Chrostowski, N. A. F. Jaeger, and D. V. Plant, "Polarization-independent mode-evolution-based coupler for the silicon-on-insulator platform," *IEEE Photonics J.* **10**, 4900410 (2018).
10. L. Liu, Q. Deng, and Z. Zhou, "Subwavelength-grating-assisted broadband polarization-independent directional coupler," *Opt. Lett.* **41**(7), 1648–1651 (2016).
11. H. Xu and Y. Shi, "Ultra-compact polarization-independent directional couplers utilizing a subwavelength structure," *Opt. Lett.* **42**(24), 5202–5205 (2017).
12. Y.-L. Wu, F.-C. Liang, H.-C. Chung, and S.-Y. Tseng, "Adiabaticity engineering in optical waveguides," *Opt. Express* **28**(20), 30117–30129 (2020).
13. Y. Fu, T. Ye, W. Tang, and T. Chu, "Efficient adiabatic silicon-on-insulator waveguide taper," *Photonics Res.* **2**(3), A41–A44 (2014).
14. A. Melikyan and P. Dong, "Adiabatic mode converters for silicon photonics: Power and polarization broadband manipulators," *APL Photonics* **4**(3), 030803 (2019).
15. H.-C. Chung, S. Martínez-Garaot, X. Chen, J. G. Muga, and S.-Y. Tseng, "Shortcuts to adiabaticity in optical waveguides," *EPL* **127**(3), 34001 (2019).
16. D. F. Siriani and J.-L. Tambasco, "Adiabatic guided wave optics—a toolbox of generalized design and optimization methods," *Opt. Express* **29**(3), 3243–3257 (2021).
17. I. K. Kim, D. U. Kim, V. H. Nguyen, S. Han, and T. J. Seok, "High-performance and compact silicon photonic 3-dB adiabatic coupler based on shortest mode transformer method," *IEEE Photonics J.* **13**(4), 1–6 (2021).
18. S. Martínez-Garaot, A. Ruschhaupt, J. Gillet, Th. Busch, and J. G. Muga, "Fast quasiadiabatic dynamics," *Phys. Rev. A* **92**(4), 043406 (2015).
19. D. Stefanatos and E. Paspalakis, "Speeding up adiabatic passage with an optimal modified Roland-Cerf protocol," *J. Phys. A Math. Theor.* **53**(11), 115304 (2020).
20. Y.-J. Hung, Z.-Y. Li, H.-C. Chung, F.-C. Liang, M.-Y. Jung, T.-H. Yen, and S.-Y. Tseng, "Mode evolution based silicon-on-insulator 3 dB coupler using the fast quasiadiabatic dynamics," *Opt. Lett.* **44**(4), 815–818 (2019).
21. C.-L. Chen, *Foundations for guided-wave optics* (John Wiley & Sons, 2007).
22. C. R. Doerr, M. Cappuzzo, E. Chen, A. Wong-Foy, L. Gomez, A. Griffin, and L. Buhl, "Bending of a planar lightwave circuit 2×2 coupler to desensitize it to wavelength, polarization, and fabrication changes," *IEEE Photonics Technol. Lett.* **17**(6), 1211–1213 (2005).

23. H.-C. Chung, K.-S. Lee, and S.-Y. Tseng, "Short and broadband silicon asymmetric Y-junction two-mode (de)multiplexer using fast quasiadiabatic dynamics," *Opt. Express* **25**(12), 13626–13634 (2017).
24. L. Chrostowski, X. Wang, J. Flueckiger, Y. Wu, Y. Wang, and S. T. Fard, "Impact of fabrication non-uniformity on chip-scale silicon photonic integrated circuits," in *Optical Fiber Communication Conference*, OSA Technical Digest (Optical Society of America, 2014), paper Th2A.37.
25. G. F. R. Chen, J. R. Ong, T. Y. L. Ang, S. T. Lim, C. E. Png, and D. T. H. Tan, "Short and broadband silicon asymmetric Y-junction two-mode (de)multiplexer using fast quasiadiabatic dynamics," *Sci. Rep.* **7**(1), 7246 (2017).